POMDPs

Ben Kraske

POMDPs

- We've been living a lie:

```
s = observe(env)
```

Types of Uncertainty

Types of Uncertainty

Alleatory

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Alleatory

Types of Uncertainty

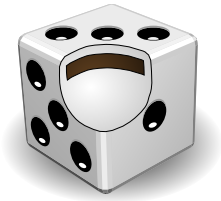
Alleatory

Epistemic (Static)

Types of Uncertainty

Alleatory

Epistemic (Static)



Types of Uncertainty

Alleatory

Epistemic (Static)

Epistemic (Dynamic)

Types of Uncertainty

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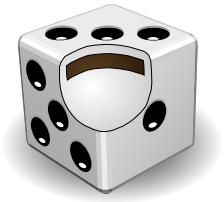
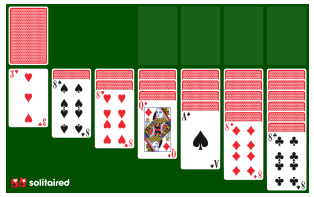
Epistemic (Static)



Epistemic (Dynamic)



Types of Uncertainty

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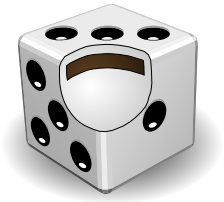
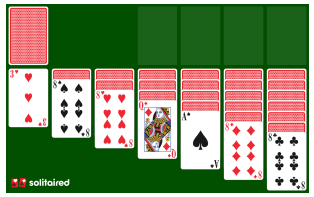


Epistemic (Dynamic)



Interaction

Types of Uncertainty

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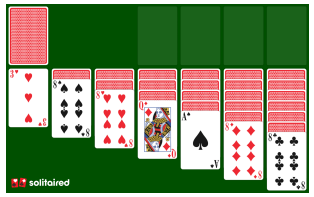

Interaction

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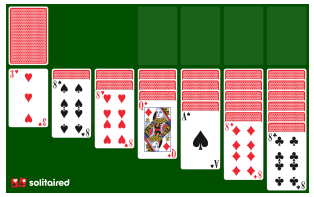
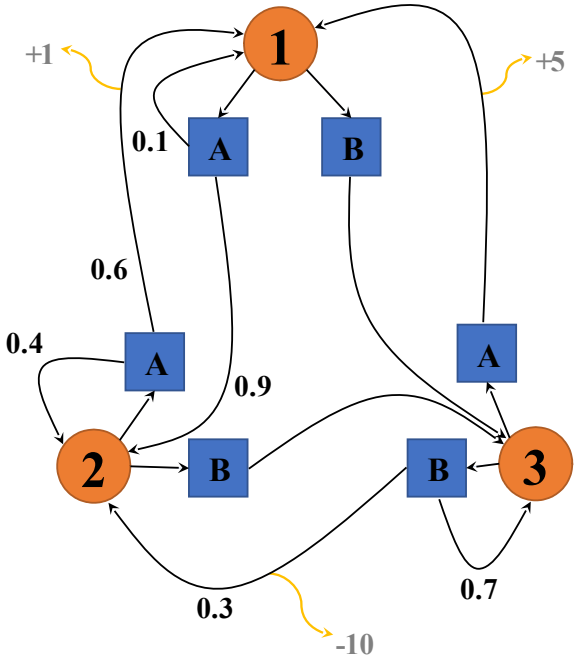


POMDPs!

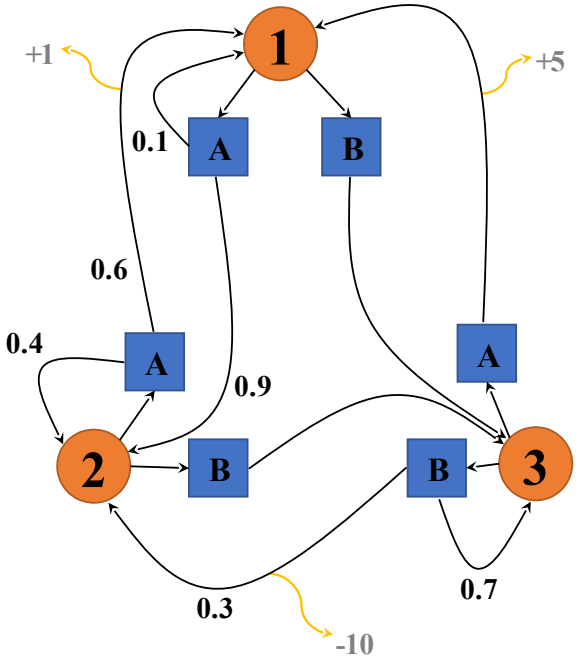
Interaction

Markov Decision Process (MDP)



- $\mathcal{S}$ - State space
- $T(s' \mid s, a)$ - Transition probability distribution

Markov Decision Process (MDP)



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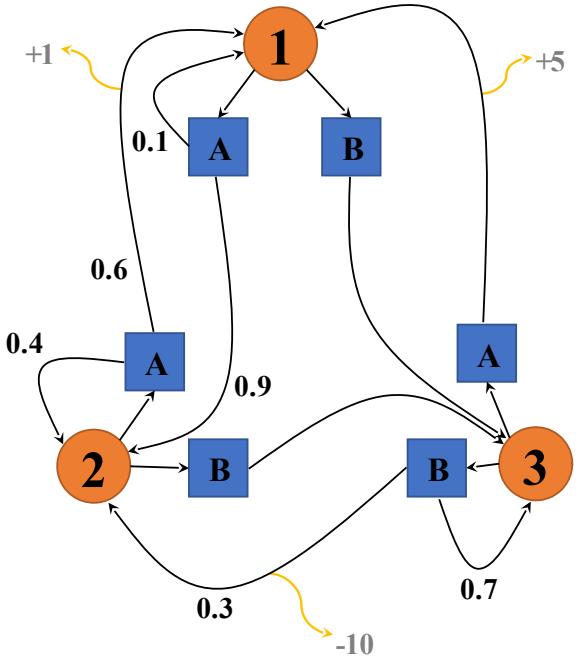

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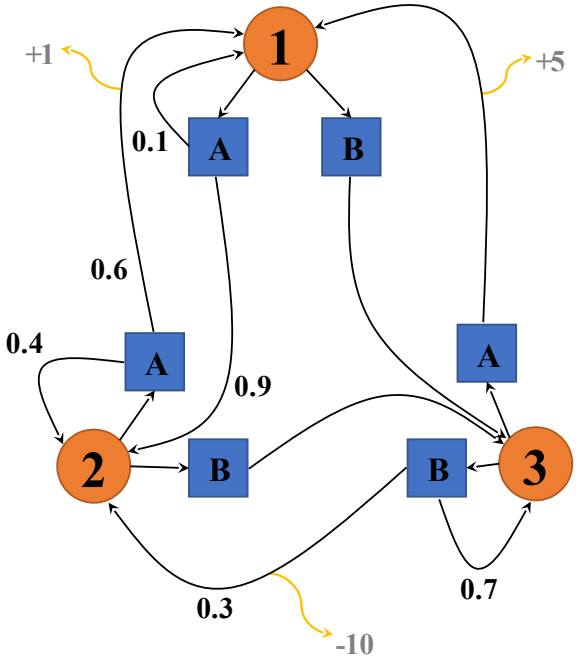

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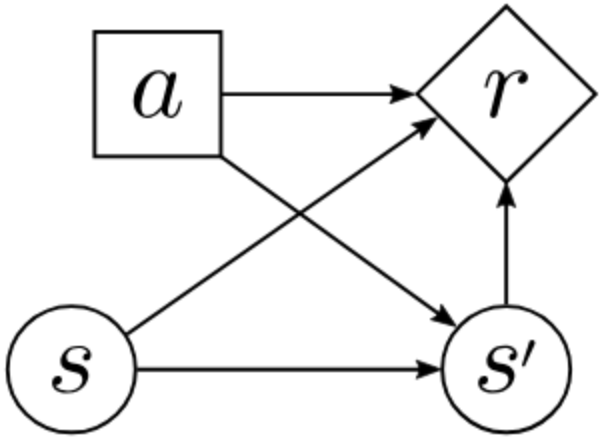

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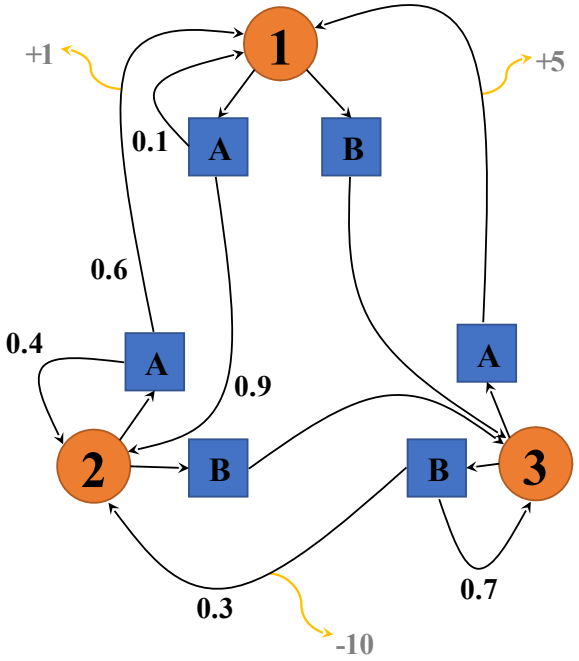

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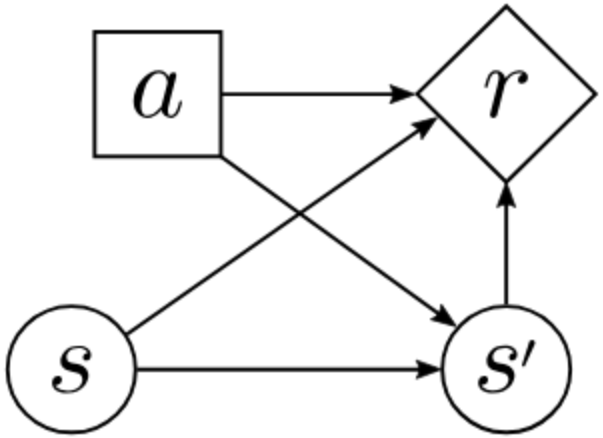
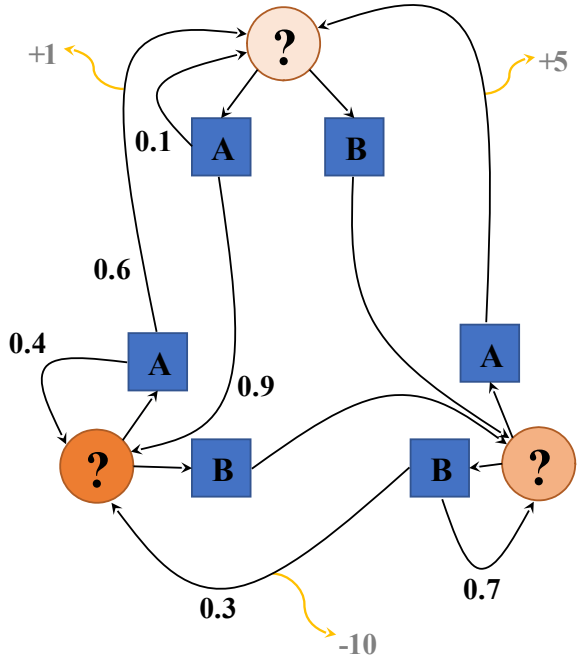


Alleatory

Partially Observable Markov Decision Process (POMDP)



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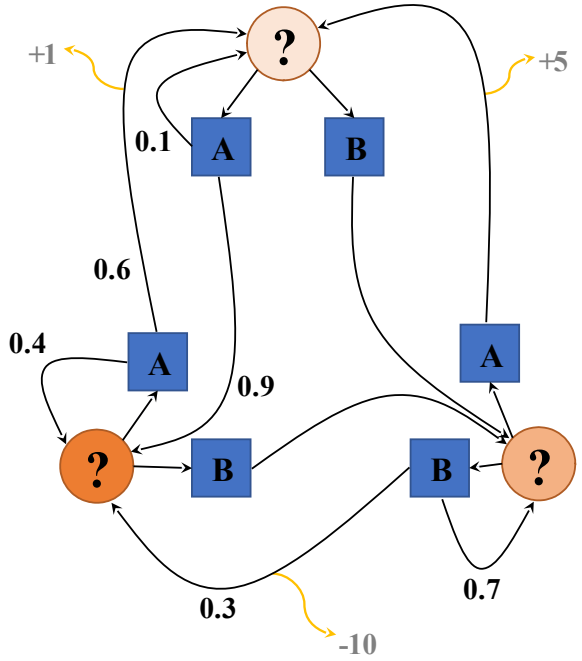

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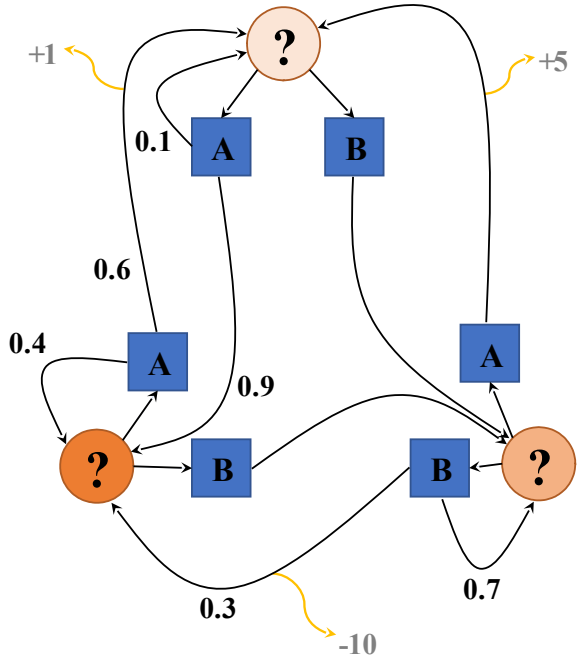

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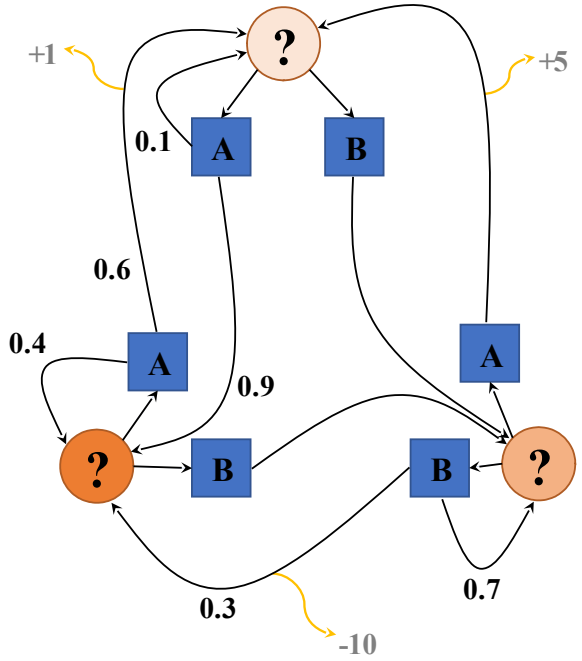

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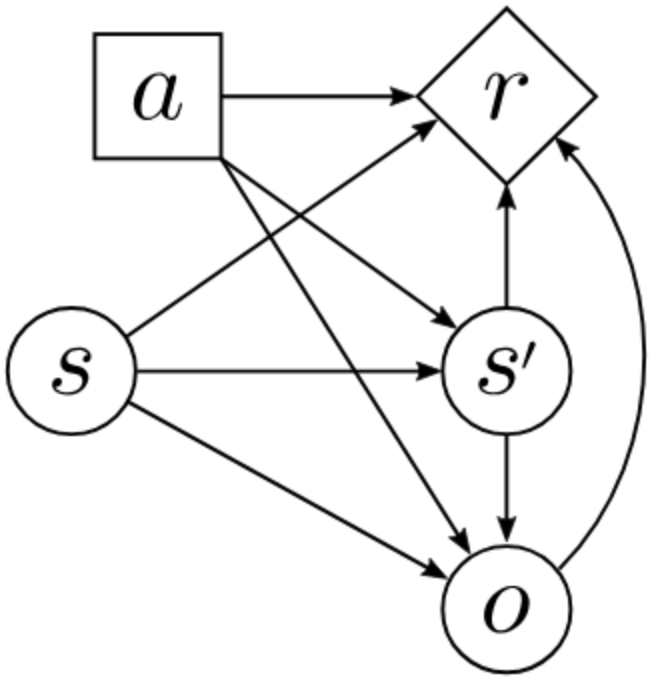

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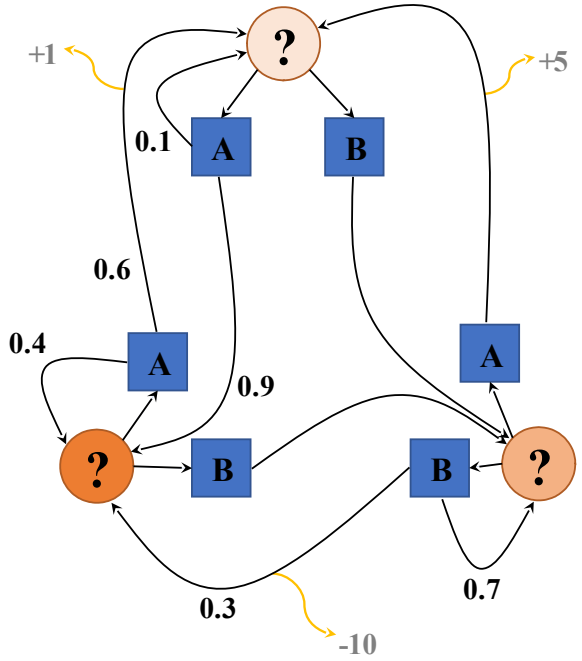

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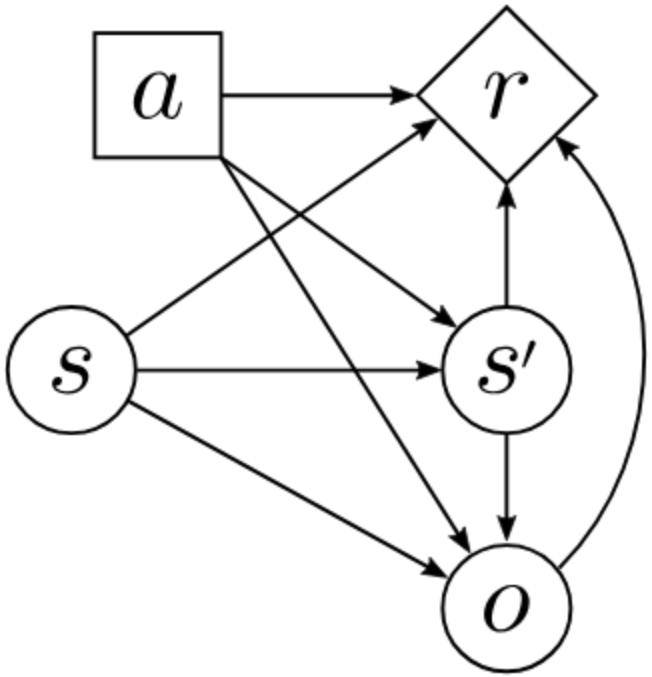

Alleatory

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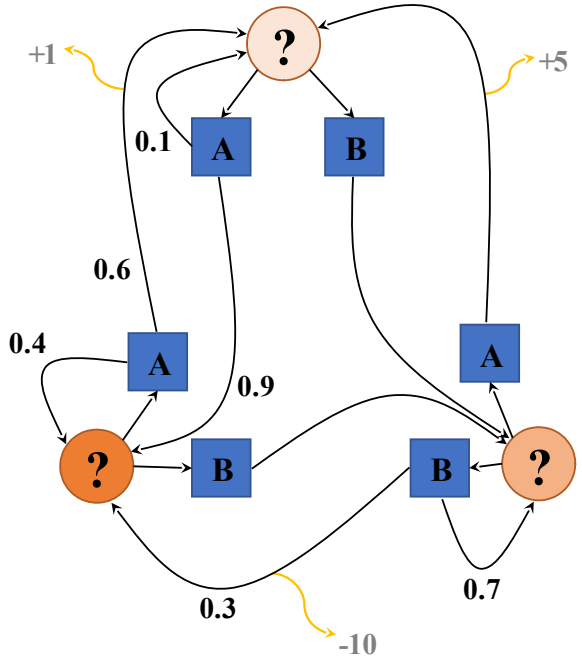

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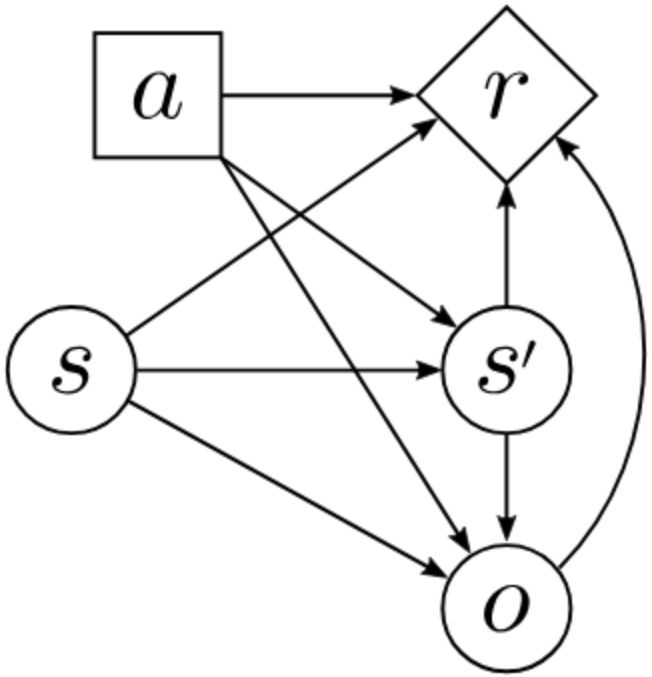

Alleatory

Epistemic (Static)

Epistemic (Dynamic)

Tiger POMDP Definition

Hidden Markov Models and Beliefs

Bayesian Belief Updates

Filtering Loop

Tiger Example

Recap